CHAPTER 13 · REFERENCE DEPTH

The Language of the Air — Phonetics, Numbers & Time

From here the book turns from electronics to the spoken language of the air. A radio frequency is useless if "five" is heard as "nine" or "to" is mistaken for "two". So aviation built its own precise dialect: a spelling alphabet where every word sounds different, exact ways to speak numbers and time, and a small set of standard words that carry one meaning each.

SYLLABUS MAP

Part III (i) Transmitting technique · transmission of letters, numbers & time · standard words & phrases · test procedures

◎ Learning objectives — by the end of this chapter you will be able to...

- Apply good transmitting technique (RSVP) and the listen-before-transmit discipline.
- Spell using the ICAO phonetic alphabet, with correct emphasis.

Join live classes for guided practice and doubt clearing.

- Transmit numbers, frequencies, levels, headings, wind and pressure correctly, and apply the number rules.
- Transmit time in UTC.
- Use the standard words and phrases with their exact single meanings.
- Carry out a radio check and report readability on the 1–5 scale.

13.1 Transmitting technique (RSVP)

13.2 The phonetic alphabet

13.3 Transmission of numbers

13.4 The number rules

13.5 Levels, headings, wind & pressure

13.6 Time

13.7 Standard words & phrases

13.8 Radio check & readability



Join live classes for guided practice and doubt clearing.

Precision in radio telephony is critical. The phonetic alphabet ensures that call signs and critical data are never misunderstood.

13.1 Transmitting technique (RSVP)

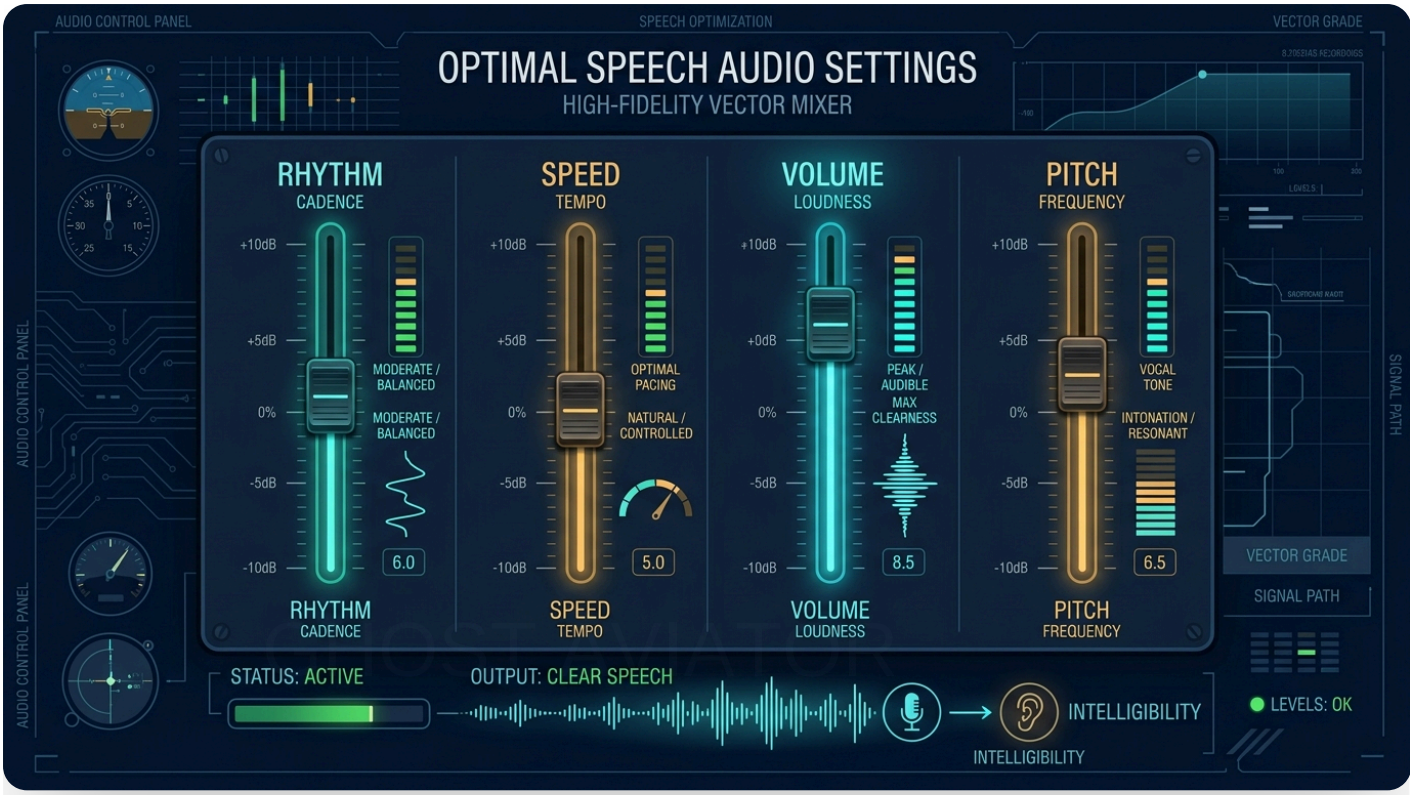
FIRST PRINCIPLES — CLARITY OVER SPEED

A transmission heard wrongly is worse than no transmission at all. The whole craft of radio is making yourself **understood the first time** on a noisy, shared channel. Before keying the microphone, **listen out** to be sure you won't block another station, then think what you will say.

RSVP — THE FOUR LEVERS OF GOOD SPEECH

- **R — Rhythm:** speak in a steady, even cadence; group the message into natural chunks.
- **S — Speed:** slightly slower than normal conversation, especially for numbers.
- **V — Volume:** constant, normal level; don't shout or trail off.
- **P — Pitch:** keep it level; a slightly higher, clear pitch carries better than a mumble.

Join live classes for guided practice and doubt clearing.



Like an audio mixer, balancing Rhythm, Speed, Volume, and Pitch is the key to perfect RT clarity.

⚠ EXAM TRAP — THE PRACTICAL

The examiner is judging exactly this. The most common practical fails are **too fast, swallowed numbers**, and forgetting to **release the PTT**. Slow down on figures and pause a beat after keying so the first word isn't clipped.

GHOST AVIATOR

13.2 The phonetic alphabet

ℹ IN PLAIN TERMS

Join live classes for guided practice and doubt clearing.

Each letter is spoken as a whole word chosen to sound unlike every other — so "B" and "D", or "M" and "N", can never be confused. The **bold syllable** is stressed.

The ICAO Phonetic Alphabet

Letter	Word	Letter	Word	Letter	Word
A	**AL**-fa	J	**JU**-li-**ETT**	S	Si-**ER**-ra
B	**BRA**-vo	K	**KI**-lo	T	**TANG**-go
C	**CHAR**-lie	L	**LI**-ma	U	**U**-ni-form
D	**DEL**-ta	M	**MIKE**	V	**VIC**-tor
E	**ECH**-o	N	No-**VEM**-ber	W	**WHIS**-key
F	**FOX**-trot	O	**OS**-car	X	**X**-ray
G	**GOLF**	P	Pa-**PA**	Y	**YANG**-key
H	Ho-**TEL**	Q	Ke-**BECK**	Z	**ZOO**-loo

GHOST AVIATOR

Join live classes for guided practice and doubt clearing.

Letter	Word	Letter	Word	Letter	Word
I	**IN**-di-a	R	**RO**-me-o		

> **⚠ EXAM TRAP — SPELLING & PRONUNCIATION**
>
> Watch the unusual spellings: **Alfa** (not Alpha), **Juliett** (double-t), **Whiskey**, **X-ray**.
>
> Pronounce the stressed syllable: **AL**-fah, **BRAH**-VOH, **CHAR**-lee, **KEY**-LOH, etc.
>
> Examiners often ask you to spell your own registration.

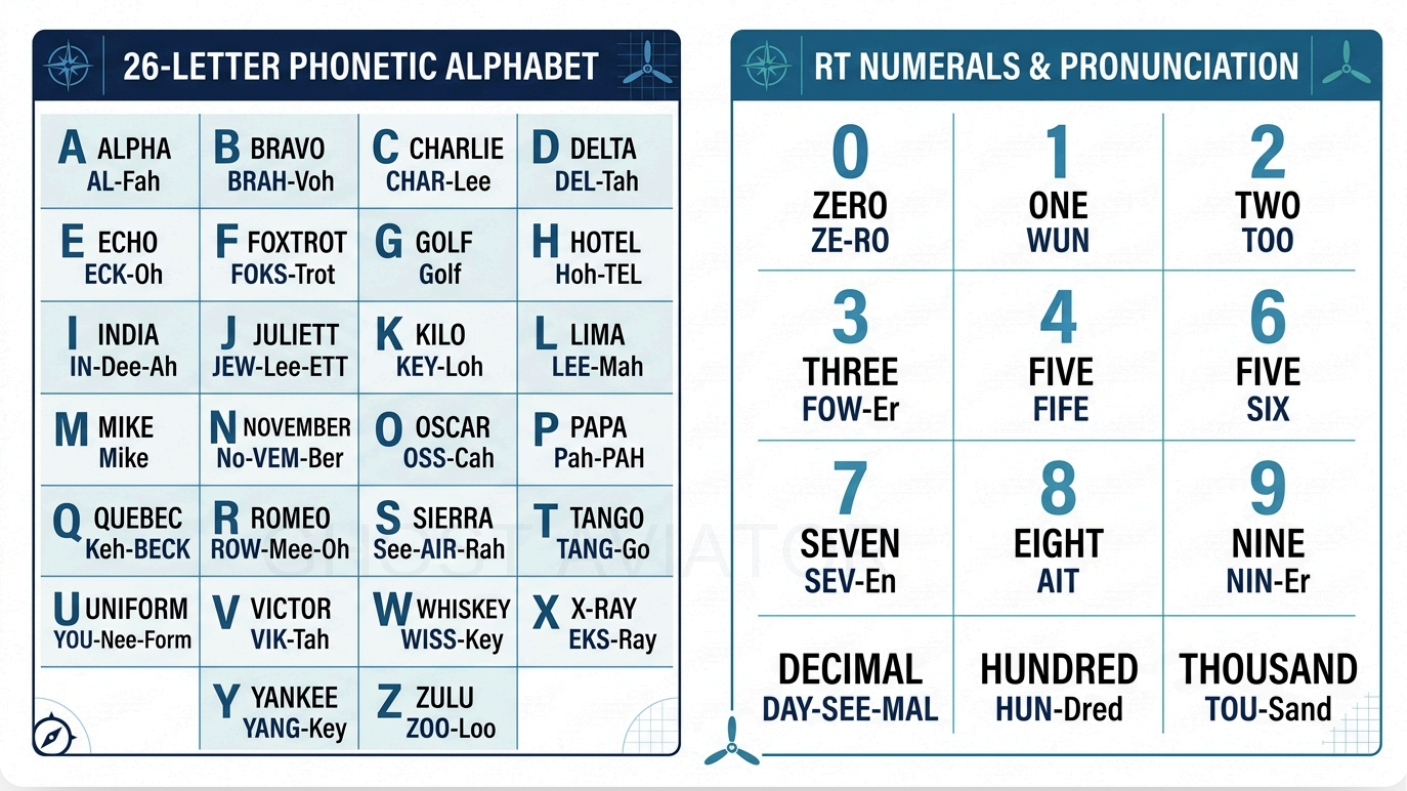


Figure 13.1 — The phonetic alphabet & number pronunciation reference card.

Join live classes for guided practice and doubt clearing.

13.3 Transmission of numbers

Pronunciation of Numerals			
Numeral	**Pronounced**	**Numeral**	**Pronounced**
0	ZE-RO	**5**	FIFE
1	WUN	**6**	SIX
2	TOO	**7**	SEV-en
3	TREE	**8**	AIT
4	FOW-er	**9**	NIN-er
Decimal	DAY-SEE-MAL	**Hundred / Thousand**	HUN-dred / TOU-SAND

GHOST AVIATOR

⚠ EXAM TRAP — THE "SAFE" PRONUNCIATIONS

Three (**TREE**), five (**FIFE**), nine (**NIN-er**) and decimal (**DAY-SEE-MAL**) are deliberately altered so they cannot be confused with similar-sounding words or other languages. Use them exactly.

Join live classes for guided practice and doubt clearing.

13.4 The number rules

THE TWO RULES

Rule 1 — digit by digit. Most numbers are transmitted by pronouncing **each digit separately**: aircraft call signs, headings, wind, pressure settings, transponder codes, frequencies, and flight levels.

Rule 2 — whole hundreds & thousands. Whole hundreds and whole thousands are transmitted by pronouncing each digit in the number of hundreds/thousands followed by **HUNDRED** or **THOUSAND** — used for altitudes, heights, cloud heights and visibility.

GHOST AVIATOR

13.5 Levels, headings, wind & pressure

Applying the Number Rules

Item	How to say it	Example
Flight level	"Flight level" + each digit	FL280 → "flight level two eight zero"
Altitude / height	Whole thousands/hundreds	3,500 ft → "three thousand five hundred feet"

Join live classes for guided practice and doubt clearing.

Item	How to say it	Example
Heading	"Heading" + 3 digits	080 → "heading zero eight zero"
Wind	Direction (3 digits) + speed	"wind two seven zero degrees, one zero knots"
Pressure (QNH)	"QNH" + each digit (units optional)	1013 → "QNH one zero one three"
Frequency	Each digit incl. "decimal"	121.500 → "one two one decimal five"

GHOST AVIATOR

WORKED EXAMPLES — NUMBERS SPOKEN CORRECTLY

Heading 270 → "heading two seven zero" (digit by digit). **Squawk 4321** → "squawk four three two one".

Altitude 2,500 ft → "two thousand five hundred". **Flight level 100** → "flight level one zero zero" (digit by digit — note the difference).

Frequency 118.500 → "one one eight decimal five". **QNH 1009** → "QNH one zero zero nine".

GHOST AVIATOR

COCKPIT REALITY

Join live classes for guided practice and doubt clearing.

A controller passing "descend flight level one zero zero, QNH one zero zero nine" expects an exact read-back. The discipline of digit-by-digit numbers is what prevents a "level 100" being flown as "level 1000".

13.6 Time

UTC — ONE CLOCK FOR THE WORLD

Aviation uses **Coordinated Universal Time (UTC)**, spoken "Zulu". Time is normally given as the **minutes of the hour** only (e.g. "at three five"), unless there is any chance of confusion, when the **full hour and minutes** are given (e.g. "one three three five"). Each digit is pronounced separately.



Join live classes for guided practice and doubt clearing.

Aviation relies on a single global time standard: Coordinated Universal Time (UTC), commonly referred to as "Zulu" time in RT.

WORKED EXAMPLE

It is 1335 UTC. Normal: "time three five". If confusion is possible (e.g. near the hour): "time one three three five".

GHOST AVIATOR

13.7 Standard words & phrases

Core Phraseology	
Word / phrase	**Single meaning**
AFFIRM	Yes
NEGATIVE	No / not correct / not available
ROGER	I have received all of your last transmission (NOT "yes")
WILCO	I understand your message and will comply
STANDBY	Wait, I will call you
SAY AGAIN	Repeat all (or the specified part) of your last transmission

GHOST AVIATOR

Join live classes for guided practice and doubt clearing.

Word / phrase	Single meaning
READ BACK	Repeat all of this message back to me exactly
CORRECTION	An error has been made; the correct version is …
DISREGARD	Ignore that transmission
CONFIRM	Have I correctly received …? / Confirm you have received …
MONITOR	Listen out on (frequency)
REQUEST	I should like to know … / I wish to obtain …
OUT	This exchange is ended; no reply expected

GHOST AVIATOR

⚠ EXAM TRAP — ROGER VS WILCO VS AFFIRM

A favourite: **ROGER** means "received", **not** "yes" and **not** "I will comply". If asked a yes/no question, answer **AFFIRM** or **NEGATIVE**; if instructed to do something, answer **WILCO**. "Over and out" is wrong — they are mutually exclusive; in modern RT you usually just end naturally.

GHOST AVIATOR

13.8 Radio check & readability

THE READABILITY SCALE (1–5)

1 Unreadable · **2** Readable now and then · **3** Readable but with difficulty · **4** Readable · **5** Perfectly readable.

TRANSCRIPT — A RADIO CHECK

A/C Delhi Tower, VT-ABC, radio check on one one eight decimal one.

ATC VT-ABC, Delhi Tower, readability five.

A/C VT-ABC, readability five also.

GHOST AVIATOR

☆ Numbers to memorise

Essential Facts for Chapter 13

Fact	Value
RSVP	Rhythm · Speed · Volume · Pitch
Altered numerals	3 = TREE · 5 = FIFE · 9 = NIN-er · decimal = DAY-SEE-MAL
Rule 1	Digit by digit: call signs, headings, wind, QNH, squawk, frequency, flight levels

GHOST AVIATOR

Fact	Value
Rule 2	Whole hundreds/thousands: altitude, height, cloud, visibility
Time	UTC (Zulu); minutes only, or full hours+minutes if confusion
ROGER / WILCO	Received / will comply (ROGER ≠ yes)
AFFIRM / NEGATIVE	Yes / No
Readability	1 unreadable → 5 perfectly readable

GHOST AVIATOR

Question bank

Part A — MCQs (click an option to check)

GHOST AVIATOR

1. In the phonetic alphabet, the letter "J" is:

Jupiter

Juliet

Juliett

Jig

GHOST AVIATOR

Answer: Juliett.

"Juliett" — spelled with a double-t.

GHOST AVIATOR

2. The numeral 9 is pronounced:

Nine

NIN-er

Niner-zero

Nin

GHOST AVIATOR

Answer: NIN-er.

9 = "NIN-er", altered to avoid confusion with "no/nein".

GHOST AVIATOR

3. The decimal point is spoken as:

Point

Dot

DAY-SEE-MAL

Comma

GHOST AVIATOR

Answer: DAY-SEE-MAL.

"Decimal" (DAY-SEE-MAL) — e.g. 118.1 = "one one eight decimal one".

GHOST AVIATOR

4. Flight level 100 is transmitted as:

"flight level one hundred"

"flight level one zero zero"

"flight level ten thousand"

"level one hundred"

GHOST AVIATOR

Answer: "flight level one zero zero".

Flight levels are spoken digit by digit: "one zero zero".

GHOST AVIATOR

5. An altitude of 3,500 ft is transmitted as:

"three five zero zero"

"three thousand five hundred"

"flight level three five"

"thirty-five hundred"

GHOST AVIATOR

Answer: "three thousand five hundred".

Altitudes use whole thousands/hundreds: "three thousand five hundred".

GHOST AVIATOR

6. A heading of 080° is spoken as:

"heading eighty"

"heading zero eight zero"

"heading eight hundred"

"heading eight zero"

GHOST AVIATOR

Answer: "heading zero eight zero".

Headings are three digits, digit by digit: "zero eight zero".

GHOST AVIATOR

7. The frequency 121.500 is transmitted as:

"one twenty-one point five"

"one two one decimal five"

"one hundred twenty-one five hundred"

"one two one point five hundred"

GHOST AVIATOR

Answer: "one two one decimal five".

Digit by digit with "decimal": "one two one decimal five".

GHOST AVIATOR

8. "ROGER" means:

Yes

I will comply

I have received all your last transmission

Standby

GHOST AVIATOR

Answer: I have received all your last transmission.

ROGER = received; it does NOT mean "yes" or "will comply".

GHOST AVIATOR

9. The correct reply to an instruction you will carry out is:

ROGER

AFFIRM

WILCO

CONFIRM

GHOST AVIATOR

Answer: WILCO.

WILCO = "I understand and will comply".

GHOST AVIATOR

Join live classes for guided practice and doubt clearing.

10. The reply meaning "yes" is:

ROGER

AFFIRM

WILCO

CORRECT

GHOST AVIATOR

Answer: AFFIRM.

AFFIRM = yes; NEGATIVE = no.

GHOST AVIATOR

11. "STANDBY" means:

Continue your approach

Wait, I will call you

Disregard

Read back

GHOST AVIATOR

Answer: Wait, I will call you.

STANDBY = wait; the controller will call you.

GHOST AVIATOR

12. Aviation time is given in:

Local time

UTC (Zulu)

Indian Standard Time

GMT+5:30

GHOST AVIATOR

Answer: UTC (Zulu).

Aviation uses UTC, spoken "Zulu".

GHOST AVIATOR

13. Normally the time is transmitted as:

The full hours and minutes always

The minutes only, unless confusion is possible

Local time with AM/PM

Seconds only

GHOST AVIATOR

Answer: The minutes only, unless confusion is possible.

Minutes only is normal; the full hour and minutes are used if confusion could arise.

GHOST AVIATOR

14. On the readability scale, "readable but with difficulty" is:

1

2

3

5

GHOST AVIATOR

Answer: 3.

3 = readable but with difficulty; 5 = perfectly readable.

GHOST AVIATOR

15. A transponder code 7000 is read as:

"seven thousand"

"seven zero zero zero"

"seventy hundred"

"flight level seven zero"

GHOST AVIATOR

Answer: "seven zero zero zero".

Squawk codes are digit by digit: "seven zero zero zero".

GHOST AVIATOR

16. The numeral 5 is pronounced:

Five

FIFE

Fiver

Fifty

GHOST AVIATOR

Answer: FIFE.

5 = "FIFE", altered for clarity.

GHOST AVIATOR

Join live classes for guided practice and doubt clearing.

17. "SAY AGAIN" requests:

A frequency change

Repetition of all or part of the last transmission

A read-back

A radio check

GHOST AVIATOR

Answer: Repetition of all or part of the last transmission.

SAY AGAIN asks the other station to repeat.

GHOST AVIATOR

18. Good transmitting technique is summarised by:

Loud and fast

RSVP — Rhythm, Speed, Volume, Pitch

Brevity only

High pitch only

GHOST AVIATOR

Answer: RSVP — Rhythm, Speed, Volume, Pitch.

RSVP — steady rhythm, controlled (slightly slower) speed, even volume, level pitch.

Part B — Oral / viva (tap to reveal model answers)

▼ ? **Spell your registration VT-ABC using the phonetic alphabet.**

GHOST AVIATOR

Model Answer:

Victor – Tango – Alfa – Bravo – Charlie.

▼ ? **How are the numerals 3, 5 and 9 pronounced, and why are they altered?**

Model Answer:

"TREE", "FIFE" and "NIN-er". They are altered so they cannot be confused with similar-sounding words or with terms in other languages.

Join live classes for guided practice and doubt clearing.

▼ ? State the two number rules and give an example of each.

Model Answer:

Rule 1 — most numbers are spoken digit by digit (heading 270 = "two seven zero"). Rule 2 — whole hundreds and thousands use HUNDRED/THOUSAND for altitude, height, cloud and visibility (2,500 ft = "two thousand five hundred").

▼ ? What is the difference between ROGER, WILCO and AFFIRM?

Model Answer:

GHOST AVIATOR

ROGER means "I have received all your last transmission"; WILCO means "I understand and will comply"; AFFIRM means "yes". ROGER does not mean yes or will comply.

▼ ? How is time transmitted in radiotelephony?

Model Answer:

In UTC (Zulu), normally as the minutes of the hour only, or as the full hours and minutes if confusion is possible, each digit spoken separately.

▼ ? What is the readability scale?

GHOST AVIATOR

Model Answer:

1 — unreadable; 2 — readable now and then; 3 — readable but with difficulty; 4 — readable; 5 — perfectly readable.

⚡ 60-SECOND REVISION CARD

RSVP: Rhythm, Speed (slower for numbers), Volume, Pitch. Listen out, press & pause, release fully.

Join live classes for guided practice and doubt clearing.

Phonetics: Alfa…Zulu (watch Alfa, Juliett, Whiskey, X-ray). Numerals: **TREE, FIFE, NIN-er, DAY-SEE-MAL**.

Rule 1 digit-by-digit (call signs, headings, wind, QNH, squawk, freq, FL); **Rule 2** hundreds/thousands (altitude/height/cloud/vis).

Time = **UTC (Zulu)**, minutes only normally. **ROGER**=received · **WILCO**=will comply · **AFFIRM/NEGATIVE**=yes/no.

Readability **1–5**; radio check on initial contact.

© Ghost Aviator GHOST AVIATOR

GHOST AVIATOR

Join live classes for guided practice and doubt clearing.